UC Berkeley

Department of Electrical Engineering and Computer Sciences

EECS 127 - EECS 227AT. Optimization Models in Engineering

Lecture Note 11

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Principal component analysis (PCA). Consider data points $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(m)} \in \mathbb{R}^n$.

- Define the average $\tilde{\boldsymbol{x}} = \frac{1}{m}(\boldsymbol{x}^{(1)} + \cdots + \boldsymbol{x}^{(m)})$.
- Define the centered data points $\tilde{\boldsymbol{x}}^{(i)} = \boldsymbol{x}^{(i)} - \tilde{\boldsymbol{x}}$, $i = 1, \ldots, m$.
- **Question:** find a direction $\boldsymbol{z}$ (of length 1) along which $\tilde{\boldsymbol{x}}^{(1)}, \ldots, \tilde{\boldsymbol{x}}^{(m)}$ have maximum variation (in the previous example that corresponded to the 45° line).
- Define line $L = \{\alpha \boldsymbol{z} \mid \alpha \in \mathbb{R}\}$.
- The projection of $\tilde{\boldsymbol{x}}^{(i)}$ on L is $\alpha_i \boldsymbol{z}$, where

$$\alpha_i = \frac{\langle \tilde{\boldsymbol{x}}^{(i)}, \boldsymbol{z} \rangle}{\underbrace{\boxed{\|\boldsymbol{z}\|}}_{=1}} = \left(\tilde{\boldsymbol{x}}^{(i)}\right)^{\mathrm{T}} \boldsymbol{z} = \boldsymbol{z}^{\mathrm{T}} \tilde{\boldsymbol{x}}^{(i)}.$$

- Define the total square variation: $\alpha_1^2 + \cdots + \alpha_m^2$.
  (Since α_i could be positive or negative, we raise it to power 2.)

$$\sum_{i=1}^{m} \alpha_i^2 = \sum_{i=1}^{m} \left(\left(\tilde{\boldsymbol{x}}^{(i)}\right)^{\mathrm{T}} \boldsymbol{z}\right)^2 = \sum_{i=1}^{m} \left(\boldsymbol{z}^{\mathrm{T}} \tilde{\boldsymbol{x}}^{(i)}\right) \left(\left(\tilde{\boldsymbol{x}}^{(i)}\right)^{\mathrm{T}} \boldsymbol{z}\right) = \boldsymbol{z}^{\mathrm{T}} \tilde{\mathbf{X}} \tilde{\mathbf{X}}^{\mathrm{T}} \boldsymbol{z},$$

  where $\tilde{\mathbf{X}} = \begin{bmatrix} \tilde{\boldsymbol{x}}^{(1)} & \cdots & \tilde{\boldsymbol{x}}^{(m)} \end{bmatrix}$ is the centered data matrix.
- We need to find the best direction for capturing the maximum data variation:

$$\max_{\|\boldsymbol{z}\|_2 = 1} \boldsymbol{z}^{\mathrm{T}} \tilde{\mathbf{X}} \tilde{\mathbf{X}}^{\mathrm{T}} \boldsymbol{z}. \tag{1}$$

- In Theorem 1 from Lecture Note 9, we showed that

$$\max_{\|\boldsymbol{w}\|_2 = 1} \boldsymbol{w}^{\mathrm{T}} \mathbf{A}^{\mathrm{T}} \mathbf{A} \boldsymbol{w} = \max_{\|\boldsymbol{w}\|_2 = 1} \|\mathbf{A}\boldsymbol{w}\|_2^2 = \|\mathbf{A}\|_2^2 = \lambda_{\max}(\mathbf{A}^{\mathrm{T}}\mathbf{A}) = \sigma_1^2(\mathbf{A}),$$

  and the optimal $\boldsymbol{w}$ is the right singular vector of $\mathbf{A}$ corresponding to the maximum singular value σ_1.
- By comparing $\boldsymbol{z}^{\mathrm{T}} \tilde{\mathbf{X}} \tilde{\mathbf{X}}^{\mathrm{T}} \boldsymbol{z}$ and $\boldsymbol{w}^{\mathrm{T}} \mathbf{A}^{\mathrm{T}} \mathbf{A} \boldsymbol{w}$, we realize that we should select $\mathbf{A}$ to be $\tilde{\mathbf{X}}^{\mathrm{T}}$.
- Let $\tilde{\mathbf{X}} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^{\mathrm{T}}$. Then, the optimal objective value of (1) is

$$\sigma_1^2\left(\tilde{\mathbf{X}}^{\mathrm{T}}\right) = \lambda_{\max}\left(\tilde{\mathbf{X}}\tilde{\mathbf{X}}^{\mathrm{T}}\right) = \lambda_{\max}\left(\tilde{\mathbf{X}}^{\mathrm{T}}\tilde{\mathbf{X}}\right) = \boxed{\sigma_1^2\left(\tilde{\mathbf{X}}\right)},$$

  and the corresponding optimal $\boldsymbol{z}$ is the right singular vector of $\tilde{\mathbf{X}}^{\mathrm{T}}$ corresponding to σ_1, which is just the left singular vector of $\tilde{\mathbf{X}}$ corresponding to σ_1, which is just $\boxed{\boldsymbol{u}^{(1)}}$.

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Example 1. Consider an example:

$$\boldsymbol{x}^{(1)} = \begin{bmatrix} 100 \\ 5 \\ 0.01 \end{bmatrix}, \quad \boldsymbol{x}^{(2)} = \begin{bmatrix} -200 \\ -2 \\ -0.05 \end{bmatrix}.$$

In this case, most of the variation is along the x-axis, thus the optimal $\boldsymbol{z} = \boldsymbol{u}^{(1)}$ should be very close to $[1, 0, 0]^{\mathrm{T}}$.
For simplicity, assume $\boldsymbol{z} = [1, 0, 0]^{\mathrm{T}}$. Now, let's remove the component of data along $\boldsymbol{z}$ to arrive at:

$$\hat{\boldsymbol{x}}^{(1)} = \begin{bmatrix} 0 \\ 5 \\ 0.01 \end{bmatrix}, \quad \hat{\boldsymbol{x}}^{(2)} = \begin{bmatrix} 0 \\ -2 \\ -0.05 \end{bmatrix}.$$

Most of the variation in the remaining data is along the direction $[0, 1, 0]^{\mathrm{T}}$, which is *orthogonal* to the previous direction $[1, 0, 0]^{\mathrm{T}}$. ■

Let's generalize Example 1:

- Original data points: $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(m)}$.
- Centered data points: $\tilde{\boldsymbol{x}}^{(1)}, \ldots, \tilde{\boldsymbol{x}}^{(m)}$ with $\sum_{i=1}^{m} \tilde{\boldsymbol{x}}^{(i)} = \boldsymbol{0}$.
- Best direction (maximum variation): $\boldsymbol{u}^{(1)}$.
- Deflated data points (remove the data components along the best direction):

$$\begin{aligned} \hat{\boldsymbol{x}}^{(i)} &= \tilde{\boldsymbol{x}}^{(i)} - \underbrace{\alpha_i \boldsymbol{u}^{(1)}}_{\text{projected point}} = \tilde{\boldsymbol{x}}^{(i)} - \frac{\langle \boldsymbol{u}^{(1)}, \tilde{\boldsymbol{x}}^{(i)} \rangle}{\underbrace{\boxed{\|\boldsymbol{u}^{(1)}\|^2}}_{=1}} \boldsymbol{u}^{(1)} \\ &= \tilde{\boldsymbol{x}}^{(i)} - \boldsymbol{u}^{(1)} \langle \boldsymbol{u}^{(1)}, \tilde{\boldsymbol{x}}^{(i)} \rangle = \tilde{\boldsymbol{x}}^{(i)} - \boldsymbol{u}^{(1)} \left(\boldsymbol{u}^{(1)}\right)^{\mathrm{T}} \tilde{\boldsymbol{x}}^{(i)} \\ &= \left(\mathbf{I}_n - \boldsymbol{u}^{(1)} \left(\boldsymbol{u}^{(1)}\right)^{\mathrm{T}}\right) \tilde{\boldsymbol{x}}^{(i)}. \end{aligned}$$

- Note that the mean of the deflated points is still zero:

$$\frac{1}{m} \sum_{i=1}^{m} \hat{\boldsymbol{x}}^{(i)} = \frac{1}{m} \left(\mathbf{I}_n - \boldsymbol{u}^{(1)} \left(\boldsymbol{u}^{(1)}\right)^{\mathrm{T}}\right) \sum_{i=1}^{m} \tilde{\boldsymbol{x}}^{(i)} = 0,$$

  so there is no mean to remove from data.

- Deflated data points matrix:

$$\hat{\mathbf{X}} = \begin{bmatrix} \hat{\boldsymbol{x}}^{(1)} & \cdots & \hat{\boldsymbol{x}}^{(m)} \end{bmatrix} = \left(\mathbf{I}_n - \boldsymbol{u}^{(1)} \left(\boldsymbol{u}^{(1)}\right)^{\mathrm{T}}\right) \tilde{\mathbf{X}}.$$

- To find the best direction along which $\hat{\boldsymbol{x}}^{(1)}, \ldots, \hat{\boldsymbol{x}}^{(m)}$ have maximum variation, we need to solve:

$$\max_{\|\boldsymbol{z}\|_2 = 1} \boldsymbol{z}^{\mathrm{T}} \hat{\mathbf{X}} \hat{\mathbf{X}}^{\mathrm{T}} \boldsymbol{z}. \tag{2}$$

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- Using the same argument as (1), we know the optimal objective value of (2) is $\sigma_1^2(\hat{\mathbf{X}})$, while the optimal direction is the left singular vector of $\hat{\mathbf{X}}$ corresponding to $\sigma_1(\hat{\mathbf{X}})$.
- Furthermore, note that

$$\begin{aligned}
\hat{\mathbf{X}} &= \left(\mathbf{I}_n - \boldsymbol{u}^{(1)}\left(\boldsymbol{u}^{(1)}\right)^{\mathrm{T}}\right)\tilde{\mathbf{X}} \\
&= \left(\mathbf{I}_n - \boldsymbol{u}^{(1)}\left(\boldsymbol{u}^{(1)}\right)^{\mathrm{T}}\right)\mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^{\mathrm{T}} \\
&= \left(\mathbf{I}_n - \boldsymbol{u}^{(1)}\left(\boldsymbol{u}^{(1)}\right)^{\mathrm{T}}\right)\left(\sum_{i=1}^{r}\sigma_i(\tilde{\mathbf{X}})\boldsymbol{u}^{(i)}\left(\boldsymbol{v}^{(i)}\right)^{\mathrm{T}}\right) \\
&= \sum_{i=1}^{r}\sigma_i(\tilde{\mathbf{X}})\boldsymbol{u}^{(i)}\left(\boldsymbol{v}^{(i)}\right)^{\mathrm{T}} - \sum_{i=1}^{r}\sigma_i(\tilde{\mathbf{X}})\boldsymbol{u}^{(1)}\underbrace{\left(\boldsymbol{u}^{(1)}\right)^{\mathrm{T}}\boldsymbol{u}^{(i)}}_{\substack{=0 \text{ if } i\neq 1,\\ =1 \text{ if } i=1}}\left(\boldsymbol{v}^{(i)}\right)^{\mathrm{T}} \\
&= \sum_{i=1}^{r}\sigma_i(\tilde{\mathbf{X}})\boldsymbol{u}^{(i)}\left(\boldsymbol{v}^{(i)}\right)^{\mathrm{T}} - \sigma_1(\tilde{\mathbf{X}})\boldsymbol{u}^{(1)}\left(\boldsymbol{v}^{(1)}\right)^{\mathrm{T}} \\
&= \sum_{i=2}^{r}\sigma_i(\tilde{\mathbf{X}})\boldsymbol{u}^{(i)}\left(\boldsymbol{v}^{(i)}\right)^{\mathrm{T}},
\end{aligned}$$

  where r is the rank of $\tilde{\mathbf{X}}$. This means that the non-zero singular values of $\hat{\mathbf{X}}$ are $\sigma_2(\tilde{\mathbf{X}}), \ldots, \sigma_r(\tilde{\mathbf{X}})$.
- Therefore, the optimal objective value of (2) is $\sigma_1^2(\hat{\mathbf{X}}) = \boxed{\sigma_2^2(\tilde{\mathbf{X}})}$, and the corresponding optimal solution of (2) is the left singular vector of $\hat{\mathbf{X}}$ for $\sigma_2(\tilde{\mathbf{X}})$, which is $\boxed{\boldsymbol{u}^{(2)}}$.
- We can continue to remove the data components along the second important direction $\boldsymbol{u}^{(2)}$ and repeat the above process.
- The obtained directions are called the *principal components*.

Definition 1 (Principal component analysis (PCA))**.** Given points $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(m)}$ in $\mathbb{R}^n$, we form the *centered* data matrix

$$\tilde{\mathbf{X}} = \begin{bmatrix}\tilde{\boldsymbol{x}}^{(1)} & \cdots & \tilde{\boldsymbol{x}}^{(m)}\end{bmatrix},$$

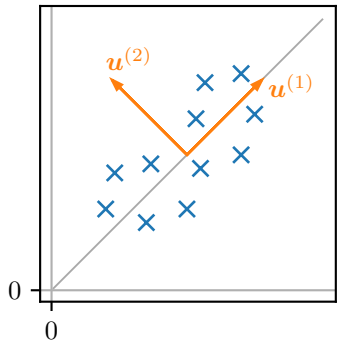
and compute its left singular vectors $\boldsymbol{u}^{(1)}, \ldots, \boldsymbol{u}^{(m)}$.
(Note: SVD for $\begin{bmatrix}\boldsymbol{x}^{(1)} & \cdots & \boldsymbol{x}^{(m)}\end{bmatrix}$ and $\tilde{\mathbf{X}}$ are not related, so we can't work on the original data without removing the mean.)
Then,

- The most data variation is along the first principal component $\boldsymbol{u}^{(1)}$.
- The next most data variation is along the second principal component $\boldsymbol{u}^{(2)}$.
- ...

Also, since $\langle \boldsymbol{u}^{(i)}, \boldsymbol{u}^{(j)} \rangle = 0$ for $i \neq j$, these principal directions are orthogonal.

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Example:



Example 2. For the US senate voting example studied earlier (Example 5 in Lecture Note 3), each $\boldsymbol{x}^{(i)} \in \mathbb{R}^m$ was the voting vector of senator i ($i \in \{1, \ldots, n\}$) on m bills. To analyze the data, we projected the centered vectors on the "average bill" direction $\boldsymbol{v} = [1, \ldots, 1]^{\mathrm{T}}$. Based on PCA, a better direction to project on would be $\boldsymbol{u}^{(1)}$, and it almost correctly identify the party (Democrat or Republican) of each senator.
Here's a Kaggle notebook that visualizes the projections on these two directions: https://www.kaggle.com/code/georgemlp2/us-senate-voting-data-2015-2016. ■

Singular values are also important in **sensitivity analysis**.

- First, consider a system of linear equations:

  Given $\mathbf{A} \in \mathbb{R}^{n \times n}$ (square) and $\boldsymbol{y} \in \mathbb{R}^n$, find $\boldsymbol{x} \in \mathbb{R}^n$ such that $\mathbf{A}\boldsymbol{x} = \boldsymbol{y}$.

- Assume $\mathbf{A}$ is invertible, then $\boldsymbol{x} = \mathbf{A}^{-1}\boldsymbol{y}$.

- What if $\boldsymbol{y}$ changes to $\boldsymbol{y} + \Delta\boldsymbol{y}$ for some small vector $\Delta\boldsymbol{y}$? In the MRI example (Lecture Note 5 and 6), the right-hand side of the system of equations is the intensity of beams and measurements are subject to noise/errors.

- Let the solution change to $\boldsymbol{x} + \Delta\boldsymbol{x}$, then we have

  $$\mathbf{A}\boldsymbol{x} = \boldsymbol{y} \quad \text{and} \quad \mathbf{A}(\boldsymbol{x} + \Delta\boldsymbol{x}) = \boldsymbol{y} + \Delta\boldsymbol{y}.$$

  Thus, $\Delta\boldsymbol{x} = \mathbf{A}^{-1}\Delta\boldsymbol{y}$.

Lemma 1. *Given matrix* $\mathbf{B}$ *and vector* $\boldsymbol{y}$*, we have*

$$\|\mathbf{B}\boldsymbol{y}\|_2 \leq \|\mathbf{B}\|_2 \cdot \|\boldsymbol{y}\|_2.$$

Proof. By definition of $\|\mathbf{B}\|_2$ (in Lecture Note 8),

$$\|\mathbf{B}\|_2 = \max_{\boldsymbol{z}} \frac{\|\mathbf{B}\boldsymbol{z}\|_2}{\|\boldsymbol{z}\|_2} \geq \frac{\|\mathbf{B}\boldsymbol{y}\|_2}{\|\boldsymbol{y}\|_2}$$

(since $\boldsymbol{y}$ is arbitrary and not optimal). Thus,

$$\|\mathbf{B}\|_2 \cdot \|\boldsymbol{y}\|_2 \geq \|\mathbf{B}\boldsymbol{y}\|_2.$$

■

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By Lemma 1, from $\Delta \boldsymbol{x} = \mathbf{A}^{-1}\Delta \boldsymbol{y}$, we conclude that

$$\|\Delta \boldsymbol{x}\|_2 \leq \|\mathbf{A}^{-1}\|_2 \cdot \|\Delta \boldsymbol{y}\|_2. \tag{3}$$

Also, from $\mathbf{A}\boldsymbol{x} = \boldsymbol{y}$, we conclude that

$$\|\boldsymbol{y}\|_2 \leq \|\mathbf{A}\|_2 \cdot \|\boldsymbol{x}\|_2. \tag{4}$$

Combining Equation (3) and Equation (4) gives us

$$\frac{\|\Delta \boldsymbol{x}\|_2}{\|\boldsymbol{x}\|_2} \leq \|\mathbf{A}\|_2\|\mathbf{A}^{-1}\|_2 \cdot \frac{\|\Delta \boldsymbol{y}\|_2}{\|\boldsymbol{y}\|_2}.$$

Here, $\|\mathbf{A}\|_2\|\mathbf{A}^{-1}\|_2$ is called the *condition number* of $\mathbf{A}$ and is shown as $\kappa(\mathbf{A})$.

Theorem 1. *In the system of equations $\boldsymbol{y} = \mathbf{A}\boldsymbol{x}$ with $\boldsymbol{x}$ being unknown and $\mathbf{A}$ being invertible, the relative change in $\boldsymbol{x}$ with regard to a relative change in $\boldsymbol{y}$ is through $\kappa(\mathbf{A})$:*

$$\frac{\|\Delta \boldsymbol{x}\|_2}{\|\boldsymbol{x}\|_2} \leq \kappa(\mathbf{A}) \cdot \frac{\|\Delta \boldsymbol{y}\|_2}{\|\boldsymbol{y}\|_2}.$$

- If $\kappa(\mathbf{A})$ is small, the solution won't change a lot with respect to a change in $\boldsymbol{y}$, meaning that the solution is not sensitive to $\boldsymbol{y}$.
- If $\kappa(\mathbf{A})$ is large, the solution *may* change a lot, meaning that it is potentially very sensitive to $\boldsymbol{y}$.
- Let

  $$\mathbf{A} = \mathbf{U}\begin{bmatrix}\sigma_1 & & 0\\ & \ddots & \\ 0 & & \sigma_n\end{bmatrix}\mathbf{V}^{\mathrm{T}}, \quad \sigma_1 \geq \sigma_2 \geq \cdots \geq \underbrace{\sigma_n > 0}_{\text{invertible}},$$

  then

  $$\mathbf{A}^{-1} = \mathbf{V}\begin{bmatrix}\frac{1}{\sigma_1} & & 0\\ & \ddots & \\ 0 & & \frac{1}{\sigma_n}\end{bmatrix}\mathbf{U}^{\mathrm{T}}.$$

  We have

  $$\begin{cases}\|\mathbf{A}\|_2 = \text{maximum singular value of } \mathbf{A} = \sigma_1,\\ \|\mathbf{A}^{-1}\|_2 = \text{maximum singular value of } \mathbf{A}^{-1} = \frac{1}{\sigma_n}\end{cases}$$
  $$\Rightarrow \kappa(\mathbf{A}) = \frac{\sigma_1}{\sigma_n} \geq 1 \qquad \text{(because } \sigma_1 \geq \sigma_n\text{)}$$
  $$\Rightarrow \kappa(\mathbf{A}) \in [1, \infty).$$

- If $\mathbf{A}$ is orthonormal, then $\mathbf{A}^{\mathrm{T}}\mathbf{A} = \mathbf{I}$. Since the singular values of $\mathbf{A}$ are the square roots of the non-zero eigenvalues of $\mathbf{A}^{\mathrm{T}}\mathbf{A}$, we have $\sigma_1 = \cdots = \sigma_n = 1$, and in this case $\kappa(\mathbf{A}) = 1$.

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- If $\kappa(\mathbf{A})$ is close to 1, then $\mathbf{A}$ is called *well-conditioned*.
- If $\kappa(\mathbf{A})$ is very large, then $\mathbf{A}$ is called *ill-conditioned*.

Consider again the system of equations $\mathbf{A}\boldsymbol{x} = \boldsymbol{y}$ with $\mathbf{A} \in \mathbb{R}^{n\times n}$ being invertible and $\boldsymbol{x}$ being unknown. Instead of perturbing $\boldsymbol{y}$, we perturb $\mathbf{A}$.
Assume $\mathbf{A}$ changes to $\mathbf{A} + \Delta\mathbf{A}$, so the model $\mathbf{A}$ is subject to noise/error:

$$\begin{aligned} &\mathbf{A}\boldsymbol{x} = \boldsymbol{y} \quad \text{and} \quad (\mathbf{A} + \Delta\mathbf{A})(\boldsymbol{x} + \Delta\boldsymbol{x}) = \boldsymbol{y} \\ \Rightarrow\ &\mathbf{A}\Delta\boldsymbol{x} = -\Delta\mathbf{A}(\boldsymbol{x} + \Delta\boldsymbol{x}) \\ \Rightarrow\ &\Delta\boldsymbol{x} = -\mathbf{A}^{-1}\Delta\mathbf{A}(\boldsymbol{x} + \Delta\boldsymbol{x}). \end{aligned}$$

Therefore,

$$\begin{aligned} \|\Delta\boldsymbol{x}\|_2 &= \|\mathbf{A}^{-1}\Delta\mathbf{A}(\boldsymbol{x} + \Delta\boldsymbol{x})\|_2 \\ &\le \|\mathbf{A}^{-1}\|_2 \cdot \|\Delta\mathbf{A}(\boldsymbol{x} + \Delta\boldsymbol{x})\|_2 \\ &\le \|\mathbf{A}^{-1}\|_2 \cdot \|\Delta\mathbf{A}\|_2 \cdot \|\boldsymbol{x} + \Delta\boldsymbol{x}\|_2 \\ &= \underbrace{\|\mathbf{A}^{-1}\|_2\|\mathbf{A}\|_2}_{\kappa(\mathbf{A})} \cdot \frac{\|\Delta\mathbf{A}\|_2}{\|\mathbf{A}\|_2} \cdot \|\boldsymbol{x} + \Delta\boldsymbol{x}\|_2. \end{aligned}$$

This gives the following theorem:

Theorem 2. *We have*

$$\frac{\|\Delta\boldsymbol{x}\|_2}{\|\boldsymbol{x} + \Delta\boldsymbol{x}\|_2} \le \kappa(\mathbf{A}) \cdot \frac{\|\Delta\mathbf{A}\|_2}{\|\mathbf{A}\|_2},$$

where the left-hand side reflects the relative change in the solution, while the right-hand side reflects the relative change in the model.

(Previously we had $\frac{\|\Delta\boldsymbol{x}\|_2}{\|\boldsymbol{x}\|_2}$. Now the denominator changed a bit.)
Therefore, $\kappa(\mathbf{A})$ shows sensitivity of the solution with respect to changes in $\boldsymbol{y}$ and $\mathbf{A}$.

- What if $\boldsymbol{y}$ and $\mathbf{A}$ simultaneously change to $\boldsymbol{y} + \Delta\boldsymbol{y}$ and $\mathbf{A} + \Delta\mathbf{A}$? This is Lemma 6.3 in textbook.
- The formula is more complicated than before, but still depends on $\kappa(\mathbf{A})$.

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